Yousif Fadhel

Electrical and Biomedical Engineer
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SKILLS

Tools - Autodesk AutoCAD, Autodesk Inventor, AGI32, AutoLUX, ArcGIS Pro, ArcGIS Online Mapping, CPS, HiCO, WaveForms, Pspice, Quartus, Keil, LTspice

Programming Languages - Arcade, Python, Java, C, C++, HTML/CSS, JavaScript, Assembly (NASM), MATLAB, Verilog

Data Structures and Algorithms - Stacks, Queues, Bubble Sort, Inheritance, Singly and Doubly Linked Lists

Lab Skills - Circuit Design, Oscilloscope understanding, PCB layout, 3D printing, FPGA Design, SLD Design

Microsoft Office Administration Tools - Microsoft Word, Microsoft Excel, Microsoft PowerPoint, Microsoft Visio

EDUCATION

McMaster University - (B.Eng.) - Electrical and Biomedical Engineering (Co-op) **Sept 2022 - Apr 2028**

Relevant Coursework: AI-Innovative Technologies (A+), Signals and Systems (A), Electromagnetics II (A-), Mechanics (A-)

EXPERIENCE

Engineering Co-Op Student - Ministry of Transportation (MTO) North York, ON M3M 1H3 **May 2025 - Sept 2026**

- Delivered roadway lighting designs and engineering packages supporting highway safety, MTO infrastructure projects and tendering
- Maintained MTO's ArcGIS-based electrical asset management system, improving accuracy of asset locations, installation records, contracts, and inspection documentation to support infrastructure planning and maintenance
- Improved RAQS application reviews by developing standardized procedures for evaluating Electrical Engineering consultant submissions
- Supported DSM compliant product selection by assessing electrical and lighting equipment against MTO, CSA, ANSI, and IES standards
- Verified roadway electrical installations through field inspections, identifying discrepancies and supporting resolution with project teams and consultants

Technical Advisor - McMaster Medical Engineering Design Team - Hamilton, ON K2K 2V6 **Oct 2024 - Present**

- Facilitated technical guidance, support, and training to 20+ students, ensuring successful implementation of engineering projects.
- Taught engineering students the fundamentals of Soldering, 3D Modelling, 3D Printing, and GitHub Navigation
- Conducted technical reviews and troubleshooting to identify and resolve engineering challenges in prototyping and design.
- Improved team efficiency through the implementation of streamlined project management tools

Team Lead - Sky Zone - Mississauga, ON L5C 2V2 **Feb 2021 - Aug 2024**

- In charge of coordinating park rotations, assuring park attractions are safely monitored, assigning closing tasks to coworkers
- Developed excellent leadership, communication and customer service skills ensuring satisfaction of all customer visits
- Operated the cashier and achieved a 30 % increase in membership sales while ensuring periodic sales goals were consistently met
- Trained 10+ new employees, improving team performance and reducing onboarding time by 20 %

PROJECTS

Highway 26 and Nottawasaga Sideroad 33 LED Retrofit (AGI32, AutoCAD, CPS) **May. 2026**

- Led electrical design coordination across disciplines for a carpool lot, roundabout, and continuous roadway lighting project.
- Optimized roadway lighting designs using **AGI32** to meet illumination requirements, minimize light trespass, and specify LED fixtures that outperform existing HPS lighting
- Created electrical layouts and wiring diagrams in **AutoCAD** to accurately document roadway lighting systems, support construction requirements, and ensure compliance with electrical design standards
- Developed detailed cost estimates, quantity sheets, and tender item lists to support project budgeting and procurement

ArcFlash Studies Report Map (ArcGIS Pro, Google Earth, Arcade) **April. 2026**

- Designed an ArcGIS feature layer that hosts distribution assembly data including their appropriate ArcFlash Report
- Implemented unique programming that flags when the DA is due for inspection based on its previous inspection date and CSA standards

Battery Voltage Monitor (IOT, Arduino, C++) - GitHub **Feb. 2025**

- Designed and assembled a lithium-ion battery monitoring system using an Arduino microcontroller, voltage divider circuitry, and cloud-based telemetry to enable real-time battery health monitoring.
- Integrated IOT by developing an Arduino **C++** program that uploaded data to the Arduino **Cloud**
- Implemented a charging module to protect the battery from Overvoltage, Overcurrent, and Short Circuiting

Automated Inhaler (Python, Inventor) - Viewer **March. 2023**

- Built a high-fidelity prototype of a wrist-attached automated inhaler
- Designed an accurately constrained cam and follower mechanism modeled on **Autodesk Inventor**
- Used a Raspberry Pi to implement a **Python** program to control the device

Personal Website (HTML,CSS,JavaScript) - GitHub **April. 2022**

- Built a website using **HTML** and **CSS** from scratch utilizing bootstrap elements and hosted on GitHub at https://yousiffadhel.github.io/
- Created a dynamic Projects section listing featured academic and independently developed projects